Specification Differences Between Products

There are the following specification differences in the RA8T2 group products, depending on chip versions.

Table 1.1 Specification Differences Depending on Chip Versions

Chapter		Specification Differences	
		Chip version A	Chip version B
section 1, Overview	section 1.3. Part Numbering	Production identification code is 0	Production identification code is 1
section 7, Option-Setting Memory	section 7.2.11. FSBLCTRL0 : FSBL Control Register 0	Prohibit to set the FSBLEN [2:0] bits to 000b	The FSBLEN[2:0] bits can be set to 000b
section 9, Clock Generation Circuit	section 9.2.36. MOSCSCR : Main Clock Oscillator Standby Control Register	Prohibit to set the MOSCSOKP bit to 1	The MOSCSOKP bit can be set to 1
	section 9.2.37. HOCOSCR : High-Speed On-Chip Oscillator Standby Control Register	Prohibit to set the HOCOSOKP bit to 1	The HOCOSOKP bit can be set to 1
section 11, Low Power Mode	section 11.7. Voltage Scaling Control	During a DVFS transition (VSCR.VSCMTSF = 1), Don’t access to the CM85's TCM and CM85's cache. Before voltage scaling operation, disable the I-cache and D-cache by setting CCR.DC and CCR.IC to 0, and then deactivate them by setting MSCR.DCACTIVE and MSCR.ICACTIVE to 0	During a DVFS transition (VSCR.VSCMTSF = 1), there is no restriction on access to the CM85's TCM and CM85's cache
section 36, EtherCAT Slave Controller (ESC)	section 36.3.3. ESCICR : ESC Interrupt Control Register	When capturing a distributed clock with the DC latch trigger (input signal from ELC or input signal from CATLATCH input terminal) selected by the ESC Interrupt Control Register, Tj must be 0 °C or higher.	When capturing a distributed clock with the DC latch trigger (input signal from ELC or input signal from CATLATCH input terminal) selected by the ESC Interrupt Control Register, there is no restriction on Tj.
section 58, MRAM	section 58.5.43. MCUVER : MCU Version Register	Value after reset is 0x1	Value after reset is 0x2
section 60, Electrical Characteristics	section 60.2.2. I/O V_{IH}, V_{IL}	TCK/SWCLK 1.62V or above V_{IH} : Min = VCC × 0.7 V_{IL} : Max = VCC × 0.3	TCK/SWCLK 1.62V or above V_{IH} : Min = VCC × 0.8 V_{IL} : Max = VCC × 0.2 ΔV_T : Min = VCC × 0.05